Sample Size for Estimating Single Mean

1. **Background**

A real estate company is evaluating a business expansion plan into a high-value residential area. The profitability and long-term financial viability of this expansion project depend heavily on property prices in the area.

To ensure the expansion is financially viable, management has determined that the average property price in the new area must exceed 1,800 K. Based on a preliminary analysis of the historical Housing dataset, the standard deviation of property prices (σ) in the population is approximately 1,870 K.

Before conducting full-scale field data collection, it is necessary to objectively determine the minimum sample size to ensure high statistical accuracy and minimize the risk of incorrect investment decisions. This report evaluates two main approaches to determining sample size.

2. **Method to to estimate the sample size**

There are two approach to estimate the sample size:

Based on desired standard error

This approach is used when a company wants to control the absolute precision of the sample mean obtained, that is, how close the sample mean is to the true population mean without directly relating it to a formal hypothesis test.

$$n = \left(\frac{\sigma}{\mathrm{SE}(\bar{y})}\right)^2$$

- n is sample size estimate
- σ is standard deviation of population
- $\mathrm{SE}(\bar{y})$ is desired standard error

Based on Power and Confidence Level

This approach is used when management wants to conduct a formal statistical inference test (hypothesis test) to make a definitive decision on whether the area meets the financial feasibility requirements or not.

$$n = \frac{\sigma^2 \left(z_{1-\frac{\alpha}{2}} + z_{1-\beta}\right)^2}{\delta^2}$$

- n is sample size estimate
- σ is standard deviation of population
- $Z1 - \alpha/2$ represents z-value under desired significance level (α)
- $Z1 - \beta$ represents z-value under the desired power
- δ The minimum difference you want to detect from the true population mean (effect size).

3. **Sample Requirement Calculation Analysis Results**

Scenario 1: Calculation Results Based on Standard Error (SE)

If the company sets a precision criterion that the standard error of the average price estimate must not be worse than 30 K, then using the parameter σ = 1,870 K, the calculation results are:

$$n = \left(\frac{1.870.000}{30.000}\right)^2$$

$$n \approx 3.885,44$$

Conclusion for Scenario 1: The company requires a minimum of 3,885 property data observations in the area to maintain the standard error of measurement below 30 K.

Scenario 2: **Calculation Results Based on Hypothesis Testing (Power and Confidence)**

If this expansion decision is based on formal statistical inference testing, management sets strict reliability standards:

- Confidence Level: 95% (meaning a significance level of α = 0.05, so the value of $z1\text{-}\alpha/2 \approx 1.96$).
- Statistical Power: 80% (meaning the probability of detecting a true effect is 80%, Type II error rate β = 0.20, so the $z1\text{-}\beta$ value is 0.84).
- Difference Effect (δ): Able to detect a difference of Rp 30,000 from the true population mean.

Using the statistical calculation function, the sample size is obtained as follows:

$$n = \frac{(1.870.000)^2 \cdot (1,96 + 0,84)^2}{(30.000)^2}$$

$$n \approx 30.461,9$$

Conclusion for Scenario 2: To conduct a robust hypothesis test with 95% accuracy and 80% detection capability, the amount of data required increases significantly to a minimum of 30,496 observations (or rounded up to 30,500 observations).

4. **Executive Summary & Managerial Recommendations**

Measurement Approach	Goal	Parameter Criteria	Minimum Sample Size
Based on Standard Error	Limiting descriptive uncertainty	SE ≤ 30.000	3.885 - 3900
Based on Hypothesis Testing	Ensuring the validity of legal business conclusions	Conf = 95%, Power = 80%, δ = 30.000	30.496 - 30.500

Strategic Recommendations for Decision-Making:

- The large difference in sample size (~3,900 vs. ~30,500) requires management to choose a data collection scenario based on the financial risk profile of this expansion project:
- Use a Target of 3,886 Samples (Scenario 1) if field data collection is subject to strict time and cost constraints, and this initial analysis is only used as an internal descriptive guide to provide a rough estimate of property prices in the new area.
- Require a Target of 30,500 Samples (Scenario 2) if the capital investment for this expansion is very large and carries a high risk of financial failure. This large sample size provides strong statistical safeguards against making costly errors (such as concluding an area is livable/expensive when it actually falls below the minimum criteria).